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Unit – 3 : Session – 11 : SLO – 2 | Circular Queue — Enqueue Operation

■ I. Algorithm to Insert an Element in a Circular Queue

- Step 1: IF FRONT = 0 AND REAR = MAX-1 (OR FRONT = REAR + 1)
- Write OVERFLOW
- Goto Step 4
- [END OF IF]
- Step 2: IF FRONT = -1 AND REAR = -1
- SET FRONT = REAR = 0
- ELSE IF REAR = MAX-1 AND FRONT != 0
- SET REAR = 0
- ELSE
- SET REAR = REAR + 1
- [END OF IF]
- Step 3: SET QUEUE[REAR] = VAL
- Step 4: EXIT

■ II. Programming Code — insert()

```
• void insert() {  
• int num;  
• printf("\n Enter the number to be inserted in the queue : ");  
• scanf("%d", &num);  
• if ((front == 0 && rear == MAX-1) || (front == rear + 1))  
• printf("\n OVERFLOW");  
• else if (front == -1 && rear == -1) {  
• front = rear = 0;  
• queue[rear] = num;  
• }  
• else if (rear == MAX-1 && front != 0) {  
• rear = 0;  
• queue[rear] = num;  
• }  
• else {  
• rear++;  
• queue[rear] = num;  
• }  
• }
```